

Colburn Analogy

Since we have an analogy between heat & mass transfer, we can use the Colburn analogy as well for external & internal flows.

$$St_x = \frac{1}{2} C_{f,x} Pr^{-2/3} \quad (\text{Heat transfer Colburn})$$

Now we need the mass transfer analog to St_x :

$$St_x = \frac{h_x}{\rho C_p U_\infty} = \frac{h_x \alpha}{k U_\infty} = \frac{q_o'' \alpha}{(T_o - T_\infty) k U_\infty}$$

For mass transfer:

$$\begin{aligned} q_o'' &\rightarrow j_o \\ T_o - T_\infty &\rightarrow C_o - C_\infty \\ \alpha &\rightarrow D \\ k &\rightarrow D \end{aligned}$$

$$\frac{q_o'' \alpha}{(T_o - T_\infty) k U_\infty} \rightarrow \frac{j_o D}{(C_o - C_\infty) D U_\infty} = \frac{h_m (C_o - C_\infty)}{(C_o - C_\infty) U_\infty} = \frac{h_m}{U_\infty}$$

$$\therefore \boxed{St_m = \frac{h_m}{U_\infty}} \equiv \text{Local mass transfer Stanton \#}.$$

Now we can say our Colburn analogy becomes:

$$\boxed{St_m \cdot Sc^{2/3} = \frac{C_{f,x}}{2}} \Rightarrow \text{Mass Transfer Colburn analogy}$$

Taking this one step further, & relating our 2 analogies:

$$\begin{aligned} St_x Pr^{2/3} &= St_m \cdot Sc^{2/3} \\ \frac{h}{\rho C_p U_\infty} \cdot Pr^{2/3} &= \frac{h_m}{U_\infty} Sc^{2/3} \end{aligned}$$

$\alpha \equiv$ mixture thermal diff.

$$\boxed{\frac{h}{h_m} = (\rho C_p) \left(\frac{\alpha}{D} \right)^{2/3} = \rho C_p Le^{2/3}} \Rightarrow \boxed{Le = \frac{Sc}{Pr} = \frac{\alpha}{D} \equiv \text{Lewis \#}}$$

So by measuring h_m , we can back calculate h !

Turbulent Flow Mass Transfer

Using our same analogy, our solution becomes trivial

$$\overline{Nu}_L = 0.037 Pr^{1/3} Re_L^{4/5} \quad (Pr > 0.5, 10^6 < Re_L < 10^8)$$

$$\Downarrow$$

$$\boxed{\overline{Sh}_L = \frac{\overline{h}_m L}{D} = 0.037 Sc^{1/3} Re_L^{4/5}} \quad (Sc > 0.5, 10^6 < Re_L < 10^8)$$

↳ Flat Plate, turbulent, incompressible

For a cylinder in turbulent cross flow:

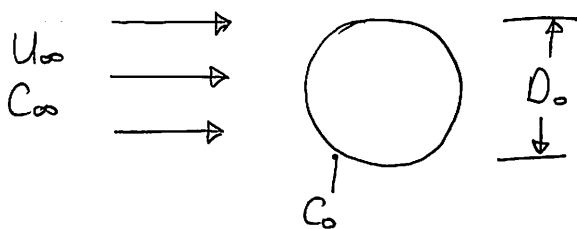
$$\boxed{\overline{Sh}_{D_o} = \frac{\overline{h}_m D_o}{D} = 0.3 + \frac{0.62 Re_{D_o}^{1/2} Sc^{1/3}}{[1 + (0.4/Sc)^{2/3}]^{1/4}} \cdot \left[1 + \left(\frac{Re_{D_o}}{282,000} \right)^{5/8} \right]^{4/5}}$$

$Re_{D_o} Sc > 0.2$

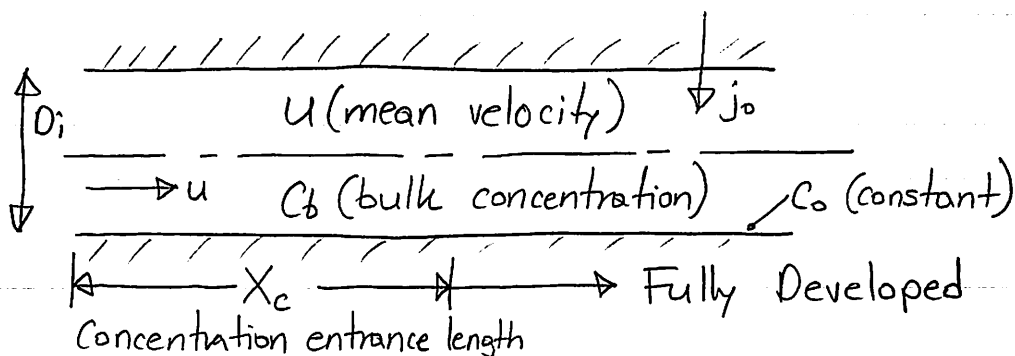
For a sphere in turbulent cross flow:

$$\boxed{\overline{Sh}_{D_o} = 2 + (0.4 Re_{D_o}^{1/2} + 0.06 Re_{D_o}^{2/3}) Sc^{0.4}} \quad (3.5 < Re_{D_o} < 7.6 \times 10^4)$$

Note, our cylinder and sphere cases look like:



Internal Forced Convection Mass Transfer



Note, our analogies still hold for internal flow:

$$\boxed{\frac{X_c}{D_h} \approx 0.04 Re_{Dh} \cdot Sc} \Rightarrow \text{Laminar Flow } (Re_{Dh} < 2300)$$

$$Re_{Dh} = \frac{UD_h}{\nu}$$

$$\boxed{C_b = \frac{1}{UA} \int_A uC dA} \equiv \text{Bulk concentration of the stream}$$

$$\boxed{Sh_{0i} = \frac{h_m D_i}{D} = 3.66} \Rightarrow \begin{array}{l} D_i = \text{tube inner diameter (pipe)} \\ D = \text{mass diffusivity} \end{array}$$

Laminar pipe flow. ($Re_{Dh} < 2300$)

For turbulent pipe flow:

$$\boxed{\frac{X_c}{D_h} \approx 10} \Rightarrow \text{Concentration entrance length in turbulent flow.}$$

Using our mass transfer analogy:

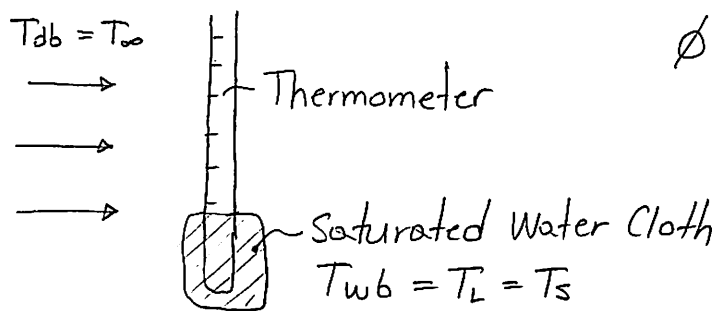
$$\boxed{Sh_{0i} = 0.023 Re_{0i}^{4/5} Sc^{1/3}} \quad (Sc > 0.5, 2 \times 10^4 < Re_{0i} < 10^6)$$

In general, we should always remember:

Heat Transfer	Mass Transfer
$h \text{ [W/m}^2 \cdot \text{K]}$	$h_m \left[\frac{\text{m}}{\text{s}} \right] = \left[\frac{\text{kg}}{\text{m}^2 \cdot \text{s} \cdot \text{kg/m}^3} \right]$
$Re \text{ (Reynolds)}$	$Re \text{ (Reynolds)}$
$Nu = \frac{hL}{k} \text{ (Nusselt)}$	$Sh = \frac{h_m L}{D} \text{ (Sherwood)}$
$Nu = f(Re, Pr)$	$Sh = f(Re, Sc)$
$Ra = \frac{g\beta\Delta TL^3}{\nu\alpha} \text{ (Rayleigh)}$	$Ram = \frac{g\Delta\rho L^3}{\mu D} \text{ (Rayleigh mass transf \#)}$

Wet Bulb Psychrometer (Note, $T_{wb} \neq T_{dew\ point}$)!

A simple device used to measure the relative humidity of air.



$\phi \equiv$ Relative Humidity

$$\phi = \frac{P_{H_2O, \infty}}{P_{H_2O, SAT}(T_{\infty})}$$

$\rightarrow$ Need a way to determine $P_{H_2O, \infty}$

Note, we defined on pg. (182) that $C_i = \rho_i \equiv$ component density

So if we do an energy balance on our psychrometer water cloth in equilibrium (evap. = convective heating)

$$Q_{EVAP} = hA(T_{db} - T_{wb}) \quad (1)$$

But we also know that evaporation $\sim h_{fg}$:

$$Q_{EVAP} = \dot{m} h_{fg} = h_m A (C_{H_2O, S} - C_{H_2O, \infty}) \cdot h_{fg} \quad (2)$$

Where $h_{fg} \equiv$ latent heat of phase change

Since $C_{H_2O, S} = \rho_{H_2O, S}$ and $C_{H_2O, \infty} = \rho_{H_2O, \infty}$

And using our ideal gas law for air/water vapor mixture

$$\rho_{H_2O, \infty} = \frac{P_{H_2O, \infty}}{R_{H_2O} \cdot T_{\infty}} \quad (3)$$

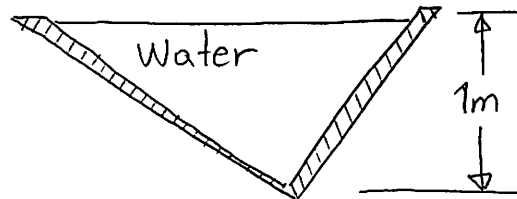
$$\rho_{H_2O, S} = \rho_{H_2O, SAT}(T_{wb}) = \frac{P_{H_2O, S}}{R_{H_2O} \cdot T_{wb}} \quad (4) \Rightarrow \text{Note the boundary condition.}$$

Rearranging & back substituting (1)-(4):

$$\left(\frac{h}{h_m} \right) \frac{(T_{\infty} - T_{wb})}{h_{fg}} = \rho_{H_2O, S} - \rho_{H_2O, \infty} \quad \Rightarrow \text{We know everything except } P_{H_2O, \infty}.$$

$$R_{H_2O} \rho C_p Le^{2/3} \cdot \frac{(T_{\infty} - T_{wb})}{h_{fg}} = \frac{P_{H_2O, S}}{T_{wb}} - \frac{P_{H_2O, \infty}}{T_{\infty}} \Rightarrow \text{Can solve explicitly for } P_{H_2O, \infty} \text{ \& } \phi. \quad (196)$$

Ex #1 | A channel 25m long & 1m deep used for storage of water. Water & surroundings are at 25°C and $RH = \phi = 50\%$.



- a) If air moves at 5m/s along the channel length, determine the rate of water loss due to evaporation.

First we need to determine the flow regime:

$$Re_L = \frac{U_{\infty} L}{\nu_{\text{air}}} = \frac{(5 \text{ m/s})(25 \text{ m})}{(15.66 \times 10^{-6} \text{ m}^2/\text{s})} = 7.98 \times 10^6 > 5 \times 10^5$$

Turbulent Flow

We can now use our analogies to solve. (Turbulent correlations)

$$\overline{Nu}_L = 0.037 Pr^{1/3} Re_L^{4/5} \quad (Pr \geq 0.5, 10^6 < Re_L < 10^8)$$

$$\overline{h} = \frac{k}{L} \cdot 0.037 Pr^{1/3} Re_L^{4/5} = \frac{(0.0267 \text{ W/m} \cdot \text{K})}{25 \text{ m}} \cdot 0.037 (0.69)^{1/3} \cdot (7.98 \times 10^6)^{4/5}$$

$$\overline{h} = 11.6 \text{ W/m}^2 \cdot \text{K}$$

We know that $\dot{m} = \overline{h}_m A (p_s - p_{\infty})$

Looking up our water vapor densities:

5m/s →

Water @ 25°C

$p_s = p_{\text{sat}}(25^\circ\text{C}) \Rightarrow p_s = 0.02282 \text{ kg/m}^3$
 $p_{\infty} = 0.5 p_{\text{sat}}(25^\circ\text{C}) = 0.01141 \text{ kg/m}^3$

Now we can solve for $\overline{h}_m$

We showed before that:

Remember, ρ & c_p are our mixture properties (air)

$$\frac{\bar{h}}{h_m} = \rho c_p Le^{2/3} ; Le = \frac{Sc}{Pr} ; h_{fg} = 2.442 \times 10^6 \text{ J/kg}$$

Remember, D is the mixture property, so D_{air} .

$$Sc = \frac{\nu}{D} \Rightarrow D_{\text{H}_2\text{O-Air}} = 22 \times 10^{-6} \text{ m}^2/\text{s} \text{ (From Table 11.1, pg. (187))}$$

$$Sc_{\text{Air}} = \frac{15.66 \times 10^{-6} \text{ m}^2/\text{s}}{22 \times 10^{-6} \text{ m}^2/\text{s}} = 0.7118$$

$$Le = \frac{Sc}{Pr} = \frac{0.7118}{0.69} = 1.0316$$

Back substituting into our analogy

$$\bar{h}_m = \frac{\bar{h}}{\rho c_p Le^{2/3}} = \frac{11.6 \text{ W/m}^2 \cdot \text{K}}{(1.177 \text{ kg/m}^3)(1005 \text{ J/kg} \cdot \text{K})(1.0316)^{2/3}}$$

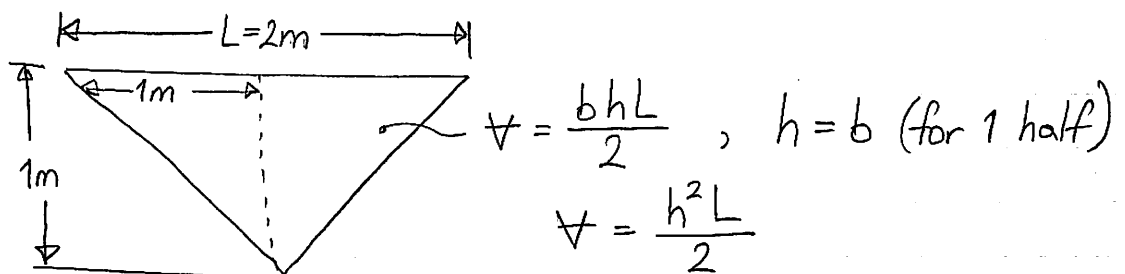
$$\bar{h}_m = 0.007675 \text{ m/s}$$

Now we can calculate $\dot{m}$ and our mass loss:

$$\dot{m} = \bar{h}_m A (p_s - p_\infty) = (0.007675 \text{ m/s})(50 \text{ m}^2)(0.02282 - 0.01141)$$

$$\dot{m} = 0.00438 \text{ kg/s} = 15.76 \text{ kg/hour}$$

- b) Obtain an expression for the rate of water depth decrease due to evaporation. How long would it take to empty the tank due to evaporation?



ρ_w since pure water in the tank.
 $\rho_w \frac{\partial V}{\partial t} = \dot{m}$

$$\frac{\dot{m}}{\rho_w} = \frac{\partial V}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} \left(\frac{h^2 L}{2} \right) = \frac{1}{2} \cdot \frac{L}{2} \frac{\partial}{\partial t} h^2 = \frac{1}{2} \cdot \frac{L}{2} \cdot 2h \cdot \frac{\partial h}{\partial t}$$

$$\int_0^H h dh = \int_0^t \frac{2\dot{m}}{\rho_w L} dt$$

$$\frac{H^2}{2} = \frac{2\dot{m}}{\rho_w L} t$$

$$t = \frac{H^2 \rho_w L}{2\dot{m}}$$

$$\Rightarrow t = \frac{(1\text{m})^2 (1000.0\text{kg/m}^3) (25\text{m})}{(2) (0.0052\text{ kg/s})}$$

$$t = 27.82 \text{ days}$$

Note, this is incorrect since we can't assume $\dot{m} = \text{constant}$.
The right way to do it is:

$$\dot{m} = \bar{h}_m (2h) L (\rho_s - \rho_\infty)$$

$$\rho \frac{\partial V}{\partial t} = \rho \frac{\partial}{\partial t} \frac{h^2}{2} \cdot L = \bar{h}_m (2hL) (\rho_s - \rho_\infty)$$

$$\rho \cancel{(2h)} \frac{1}{2} \frac{\partial h}{\partial t} \cdot \cancel{L} = \bar{h}_m (\cancel{2h} \cancel{L}) (\rho_s - \rho_\infty)$$

$$\frac{dh}{dt} = \frac{2\bar{h}_m (\rho_s - \rho_\infty)}{\rho}$$

$$h(t) = \frac{2\bar{h}_m (\rho_s - \rho_\infty)}{\rho} \cdot t + C$$

$$h(0) = H \Rightarrow C$$

$$h(t) = H + \frac{2\bar{h}_m (\rho_s - \rho_\infty)}{\rho} t \Rightarrow \text{Let } h = 0 \text{ (when dry)}$$

$$0 = H + \frac{2\bar{h}_m (\rho_s - \rho_\infty)}{\rho} t$$

$$t = 101.23 \text{ days}$$